1. 17 is related to 21 following certain logic. Following the same logic, 22 is related to 36. To which of the following numbers is 29 related, following the same logic? (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding /deleting/ multiplying etc., to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

$17 \rightarrow 21$

$22 \rightarrow 36$

$29 \rightarrow 57$

$+10 \times 3$

(a) 58 (b) 49

(c) 46 (d) 57

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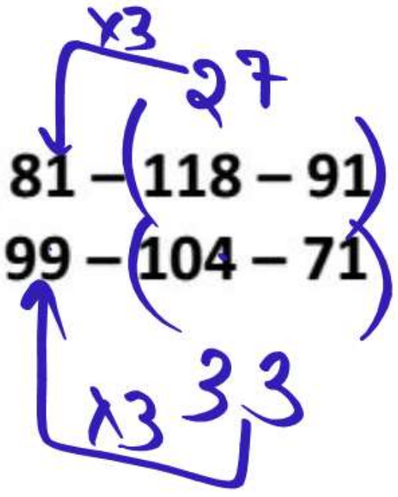
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2. Three of the following four options are alike in a certain way and thus form a group. Which is the one that does NOT belong to that group?

(NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying etc. to 13 can be perfor-med. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) Odd one out

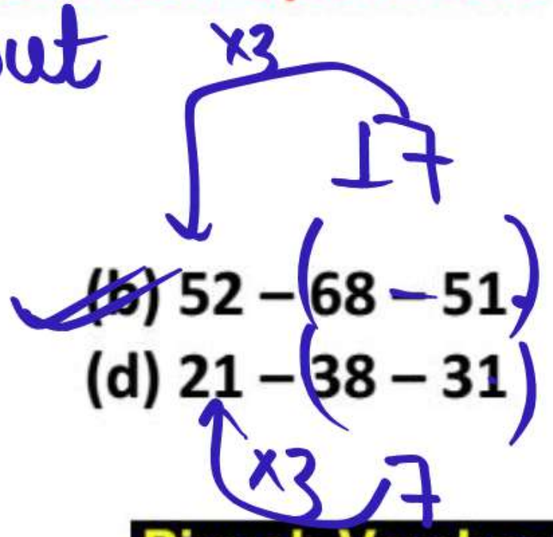
(a) 81 – 118 – 91

(c) 99 – 104 – 71



~~(b)~~ 52 – 68 – 51

(d) 21 – 38 – 31



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3. 62 is related to 79 following certain logic. Following the same logic, 98 is related to 119. To which of the following numbers is 14 related, following the same logic? (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding/deleting/multiplying etc., to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

8^2-2 62 → 79 9^2-2

10^2-2 98 → 119 11^2-2

4^2-2 14 → 5^2-2

(a) 24
(b) 23
(c) 19
(d) 21

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4. Select the set in which the numbers are related in the same way as are the numbers of the following sets.

(34, 47, 62)

(119, 142, 167)

(NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding /deleting /multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed)

(a) (27, 38, 51)

(b) (48, 65, 82)

(c) (64, 78, 99)

(d) (79, 98, 119)

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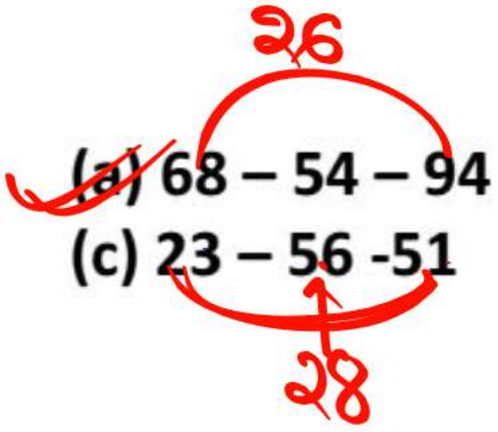
5. Three of the following four options are alike in a certain way and thus form a group. Which is the one that does NOT belong to that group?
(NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/ multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

Odd one out

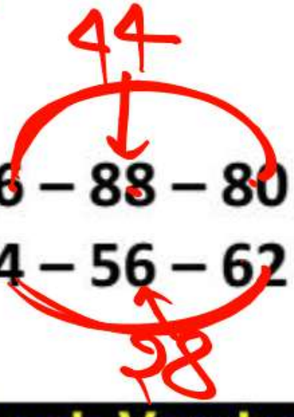
$a(c-a) = b$

$c-a = b/2$

$c = a + \frac{b}{2}$



(a) 68 – 54 – 94
(c) 23 – 56 -51



(b) 36 – 88 – 80
(d) 34 – 56 – 62

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6. Select the option in which the numbers share the same relationship as that shared by the given pair of numbers.(NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding/deleting/multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

$\times 9 + 100$

17 : 253

19 : 271

$\times 9 + 100$

$a \times 9 + 100 = b$

$9a + 100 = b$

$9a = b + 100$

$a = \frac{b + 100}{9}$

(a) 21 : 289

(b) 23 : 207

(c) 31 : 400

(d) 32 : 387

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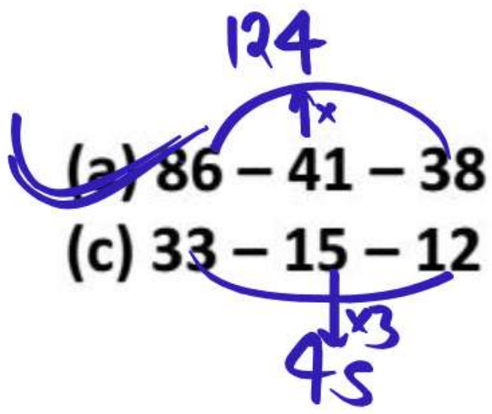
7. Three of the following four options are alike in a certain way and thus form a group. Which is the one that does NOT belong to that group?

(NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

Odd one out

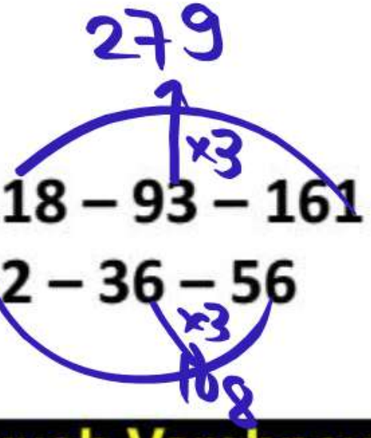
$\frac{a+c}{3} = b$

$a + c = 3b$



(a) 86 – 41 – 38

(c) 33 – 15 – 12



(b) 118 – 93 – 161

(d) 52 – 36 – 56

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8. 12 is related to 235 by certain logic. Following the same logic, 14 is related to 287. To which of the following is 18 related, following the same logic? (NOTE : Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding /deleting /multiplying etc., to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

$= -9+100$

$+91$

324

91

144

91

196

91

$12 \rightarrow 235 \ (12^2+91)$

$14 \rightarrow 287 \ (14^2+91)$

$18 \rightarrow ?$

(a) 512

(c) 415

(b) 320 (18^2+91)

(d) 438

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9. Three of the following four number-pairs are alike in a certain way and thus form a group. Which number-pair does NOT belong to that group?
(NOTE: The relation should be found without breaking down the numbers into its constituent digits)

(a) 83, 97

(b) 73, 79

(c) 61, 71

(d) 31, 41

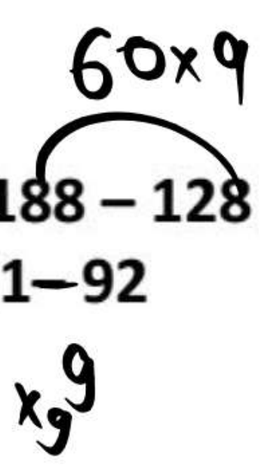
PIYUSH VARSHNEY

10. Three of the following four options are alike in a certain way and thus form a group. Which is the one that does NOT belong to that group?
(NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

$9(b-c)=a$

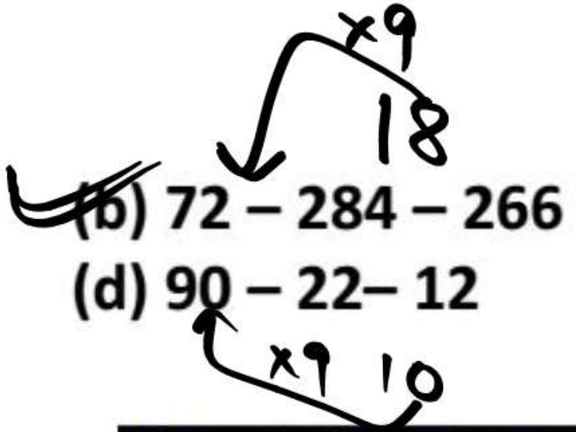
$b-c=9a$

$9a+c=b$



(a) 540 = 188 – 128

(c) 81– 101–92



(b) 72 – 284 – 266

(d) 90 – 22– 12

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11. Three of the following four options are alike in a certain way and thus form a group. Which is the one that does NOT belong to that group?
(NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/ multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) odd one out

$a = b^2 + c^3$

$a - b^2 = c^3$

$a - c^3 = b^2$

(a) $360 = 12 - 6$ $\quad 144+216$

(b) $82 - 8 - 2$ $\quad \neq 64+8$

(c) $63 - 6 - 3$ $\quad 63 = 36+27$

(d) $164 - 10 - 4$ $\quad 164 = 100+64$

12. Three of the following numbers are alike in a certain way and one is different. Pick the odd one out.
(NOTE : Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding /deleting /multiplying etc., to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) find one out

$11(a+b)=c$

$11a+11b=c$

$a+b=\frac{c}{11}$

19×11

(a) 12 + 7 – 209

~~(c)~~ 23 + 8 – 343

31×11

30×11

(b) 19 + 11 – 330

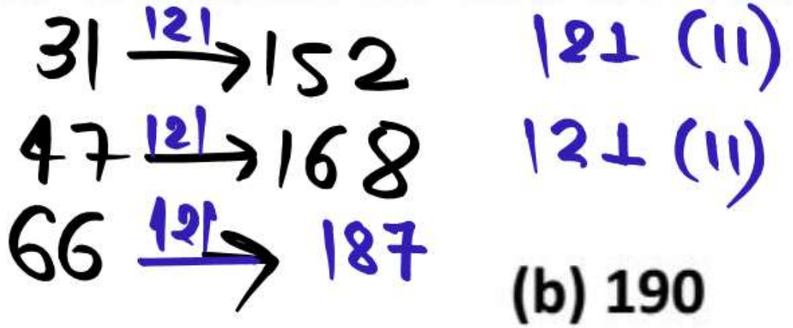
(d) 15 + 9 – 264

24×11

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13. 31 is related to 152 following certain logic. Following the same logic, 47 is related to 168. To which of the following numbers is 66 related, following the same logic? (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding /deleting/ multiplying etc., to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)



(a) 180
(b) 190
(c) 185
(d) 187

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14. The second number in the given number-pairs is obtained by performing certain mathematical operation(s) on the first number. The same operation(s) is/are followed in all the number-pairs, except one. Find that odd number-pair.(NOTE: The relation should be found without breaking down the numbers into its constituent digits)

(a) 784 : 961

(b) 1764 : 1849

(c) 1156 : 1225

(d) 1521 : 1600

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15. Which of the following interchanges of numbers (not digits) would make the given equation correct?

$$2^{57-33\div11\times13-2} + 3^{64-56\div8\times4-44} = 35$$

$2^{57-44\div11\times13-2} + 3^{64-56\div8\times4-33} = 35$

$2^3 + 3^3 = 8+27$

(a) 2 and 4 ×
(b) 56 and 57 ×
(c) 57 and 64 ×
(d) 33 and 44

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